7.7 Hyperbolic Functions

The hyperbolic functions are formed by taking combinations of the two exponential functions e^x and e^{-x}.

Recall that we say a function $f(x)$ is an **even** function if $f(-x) = f(x)$,

and that it is an **odd** function if $f(-x) = -f(x)$.

Given any function $f(x)$, we can write it as a sum of even and odd functions:

$$f(x) = \frac{f(x) + f(-x)}{2} + \frac{f(x) - f(-x)}{2}.$$

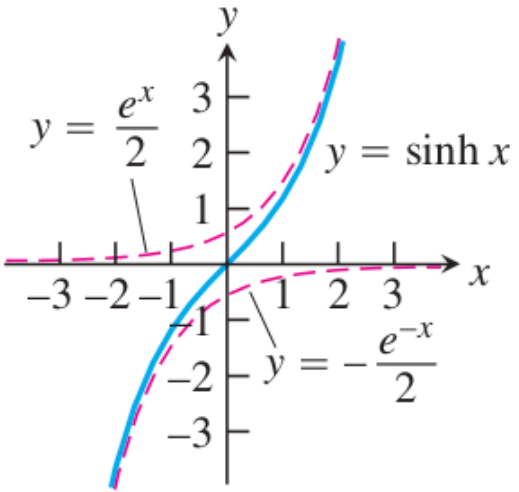
So if we let $f(x) = e^x$, we get $e^x = \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2}$.

Definitions

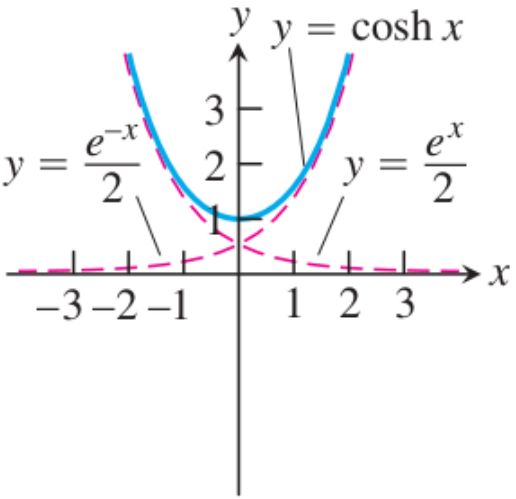
$$e^x = \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2}$$

Here we define

$$\sinh x = \frac{e^x - e^{-x}}{2} \text{ (odd)} \quad \& \quad \cosh x = \frac{e^x + e^{-x}}{2} \text{ (even).}$$

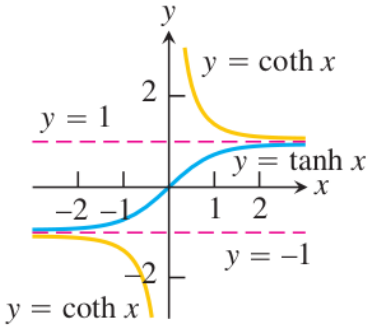


(a)

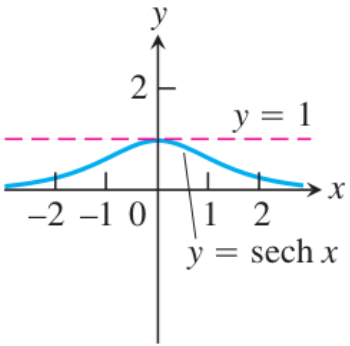


(b)

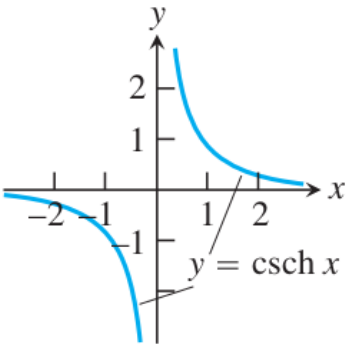
The graphs of the other hyperbolic functions.



(c)



(d)



(e)

Some identities for hyperbolic functions:

Show that

1. $\cosh^2 x - \sinh^2 x = 1$
2. $\sinh 2x = 2 \sinh x \cosh x$
3. $\cosh 2x = \cosh^2 x + \sinh^2 x$

Identities for hyperbolic functions:

- $\cosh^2 x - \sinh^2 x = 1$
- $\sinh 2x = 2 \sinh x \cosh x$
- $\cosh 2x = \cosh^2 x + \sinh^2 x$
- $\cosh^2 x = \frac{\cosh 2x + 1}{2}$
- $\sinh^2 x = \frac{\cosh 2x - 1}{2}$
- $\tanh^2 x = 1 - \operatorname{sech}^2 x$
- $\coth^2 x = 1 + \operatorname{csch}^2 x$

Derivatives of hyperbolic functions

- $\frac{d}{dx}(\sinh u) = \cosh u \frac{du}{dx}$

- $\frac{d}{dx}(\text{csch} u) = -\text{csch} u \coth u \frac{du}{dx}$

Example:

a. $\frac{d}{dx}(\cosh\sqrt{1+x^2})$

b. $\frac{d}{dx}(e^x \sinh x)$

c. $\frac{d}{dx}[(x^2-1)\operatorname{csch}(\ln x)]$

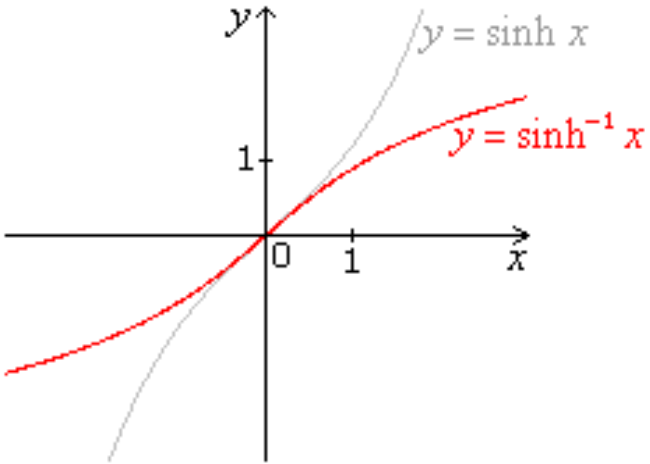
d. $\int \coth 5x \, dx$

e. $\int_{0}^{\ln 2} 4e^{x} \sinh x \, dx$

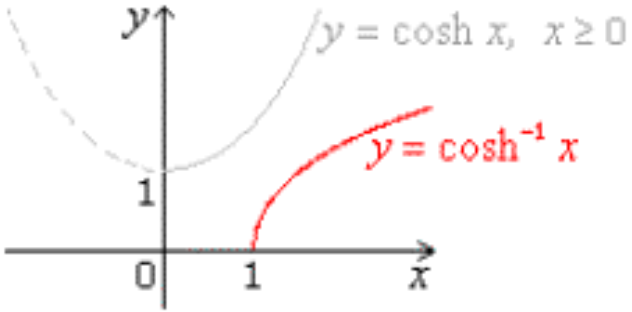
Inverse Hyperbolic Functions

The inverses of the six basic hyperbolic functions are very useful in integration.

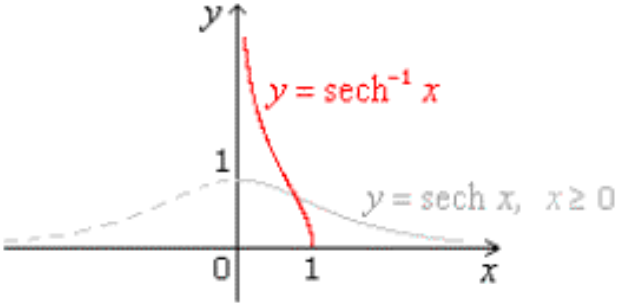
$f(x) = \sinh x$ is one-to-one.



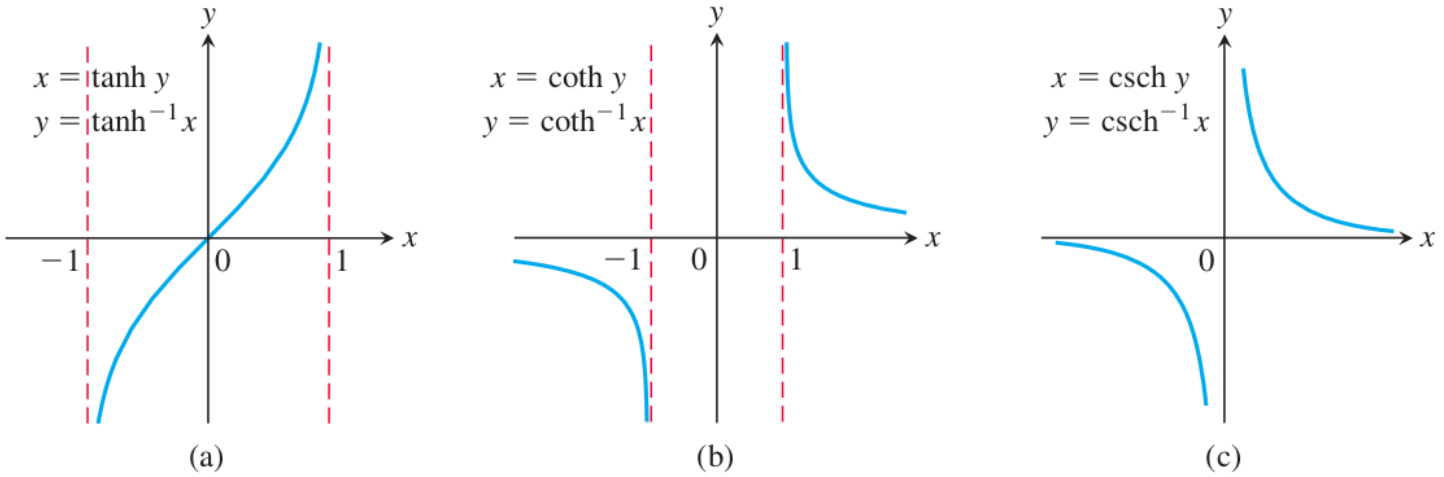
$f(x) = \cosh x$ is not one-to-one, but the restricted function $f(x) = \cosh x,\ x \in [0, \infty)$ is one-to-one.



$f(x) =$sech x is not one-to-one, but the restricted function $f(x) =$sech $x,\ x \in [0, \infty)$ is one-to-one.



The hyperbolic tangent, cotangent, and cosecant are one-to-one on their domains and therefore have inverses. (Page 441, Fig. 7.33)



Remark: 1. The number $y = \sinh^{-1} x$ is the number in $(-\infty, \infty)$ for which $\sinh y = x$.

2. Caution that $\sinh^{-1} x \neq \frac{1}{\sinh x}$.

Identities for Inverse Hyperbolic Functions

- $\operatorname{sech}^{-1} x = \cosh^{-1} \frac{1}{x},\ x \in (0, 1]$. (*)
- $\operatorname{csch}^{-1} x = \sinh^{-1} \frac{1}{x},\ x \in (-\infty, 0) \cup (0, \infty)$.
- $\coth^{-1} x = \tanh^{-1} \frac{1}{x},\ x \in (-\infty, 1) \cup (1, \infty)$.

Derivatives of inverse hyperbolic functions

$$\frac{d}{dx}(\cosh^{-1}x) = \frac{1}{\sqrt{x^2-1}},\ x > 1$$

Show that $\frac{d}{dx}(\sinh^{-1}x) = \frac{1}{\sqrt{1+x^2}}\ \forall\, x \in \mathbb{R}$.

Show that $\frac{d}{dx}(\tanh^{-1}x) = \frac{1}{1-x^2},\ |x| < 1.$

Show that $\frac{d}{dx}(\coth^{-1}x) = \frac{1}{1-x^2},\ |x| > 1.$

Example:

a. $\frac{d}{dx}(\tanh^{-1}(\sin 2x)) =$

b. $\frac{d}{dx}(\cosh^{-1}(x^2 + 1) =$

Derivatives of inverse hyperbolic functions

1. $\frac{d(\sinh^{-1}u)}{dx} = \frac{1}{\sqrt{1+u^2}}\frac{du}{dx}, \ u \in \mathbb{R}$

2. $\frac{d(\cosh^{-1}u)}{dx} = \frac{1}{\sqrt{u^2-1}}\frac{du}{dx}, \ u > 1$

3. $\frac{d(\tanh^{-1}u)}{dx} = \frac{1}{1-u^2}\frac{du}{dx}, \ |u| < 1$

4. $\frac{d(\coth^{-1}u)}{dx} = \frac{1}{1-u^2}\frac{du}{dx}, \ |u| > 1$

5. $\frac{d(sech^{-1}u)}{dx} = -\frac{1}{u\sqrt{1-u^2}}\frac{du}{dx}, \ 0 < u < 1$

6. $\frac{d(csch^{-1}u)}{dx} = -\frac{1}{|u|\sqrt{1+u^2}}\frac{du}{dx}, \ u \neq 0$

Integrals leading to inverse hyperbolic functions

1. $\int \frac{du}{\sqrt{a^2+u^2}} = \sinh^{-1}(\frac{u}{a}) + C,\ a > 0.$

2. $\int \frac{du}{\sqrt{u^2-a^2}} = \cosh^{-1}(\frac{u}{a}) + C,\ u > a > 0.$

3. $\int \frac{du}{a^2-u^2} = \frac{1}{a}\tanh^{-1}(\frac{u}{a}) + C,$ if $u^2 < a^2.$

4. $\int \frac{du}{a^2-u^2} = \frac{1}{a}\coth^{-1}(\frac{u}{a}) + C,$ if $u^2 > a^2.$

5. $\int \frac{du}{u\sqrt{a^2-u^2}} = -\frac{1}{a}\text{sech}^{-1}(\frac{u}{a}) + C, 0 < u < a.$

6. $\int \frac{du}{u\sqrt{a^2+u^2}} = -\frac{1}{a}\text{csch}^{-1}|\frac{u}{a}| + C, u \neq 0$ and $a > 0.$

Example: $\int \frac{2dx}{\sqrt{3+4x^2}} =$